

LUNCH AND LEARN • FIVE PRACTICES • 60 MINUTES

Agentic engineering

Coding agents are fast. These five practices make them reliable.

THE ART OF THE POSSIBLE

Agents can now take a ticket all the way to review.

STEP 01

Plan

Reads the ticket and plans the change.

STEP 02

Build

Writes code and tests, opens a pull request.

STEP 03

Fix

Fixes whatever breaks the build.

STEP 04

Report

Notes how sure it is, and why.

STEP 05

Hand off

Waits for approval, leaves notes.

SO WHAT

The bottleneck has moved from writing the code to trusting the result.

THE ONE RULE • USE AGENTS ANY WAY YOU LIKE

Reality must match the claim

AI raises your floor, but only you can raise your ceiling.

THE FLOOR

Anyone can now produce working-looking code in minutes. That part got cheap, for everyone.



THE CEILING

Knowing what to build, spotting what is wrong, deciding when it is good enough. That part is still yours.

SO WHAT

Everyone has a camera and a crew now. Not everyone makes a film.

YOUR ROLE • THE DIRECTOR

Direct the work: watch the take, name the problem, give one note.

STEP 01

Watch the take

Read the actual change and run it before you react.

STEP 02

Name the problem

Say exactly what is off: "the retry hides the real error."

STEP 03

Give one note

Say it once, clearly, then let the agent work.

KEY INSIGHT

A precise note gets a fix. A vague one gets a guess.

Clear and specific beats polite, rude or loud.

- | | | |
|----|-----------------------|--|
| 01 | Say what to do | "Use early returns" works better than a list of things to avoid. |
| 02 | Say why it matters | "This runs in checkout, so a wrong total costs money." |
| 03 | Skip the shouting | "CRITICAL" and "MUST" can make newer models overreact. |
| 04 | Keep the tone neutral | Studies on politeness disagree, so don't spend effort on it. |

OUR VIEW

Tone moves results a little. Missing information moves them a lot.

Set the bar before the first take, and don't settle for the first pass.

SETTLE

"It compiles and the tests pass." The agent stops at the first green result, and so do you.



SET THE BAR

Decide what done means before work starts: it runs where users run it, someone looked at it, the docs match.

SO WHAT

Passing is the floor. You set the bar.

YOUR ROLE • DECLARE THE OUTCOME

Declare what you want, and let the agent work out how.

TELLS IT HOW

1. Open orders.csv
2. Loop over each row
3. Skip rows where status is empty
4. Convert the date to ISO format
5. If the total is over 1000, flag it
6. Write each row to the database

DECLARES WHAT

```
order:  
  id: text, unique  
  date: ISO 8601 date  
  status: pending | paid | refunded  
  total: decimal, never negative  
rules:  
  - totals over 1000 need review  
  - never drop a row silently  
done when:  
  - every row loads or is reported
```

KEY INSIGHT

The schema and the checks outlive the code the agent writes.

YOUR ROLE • DECLARE THE OUTCOME

Declarative works when the spec leaves nothing to guess.

01	Define the shape	A schema or types: what the data is, and what it can never be.
02	Name the constraints	What must never happen, and which tradeoffs you accept.
03	Say how you'll know	Checks that pass only when the outcome is right.
04	Stay imperative where order matters	Migrations, rollouts and cut-overs: spell out the steps.
05	Read the how anyway	The agent's code is still yours, including its speed and safety.

Lead, follow, or get out of the way.

Lead

When the goal is unclear, set the direction and say why.

Follow

When the agent knows the ground better, let it propose a plan.

Get out of the way

When limits and checks are in place, let it run.

RECOMMENDATION

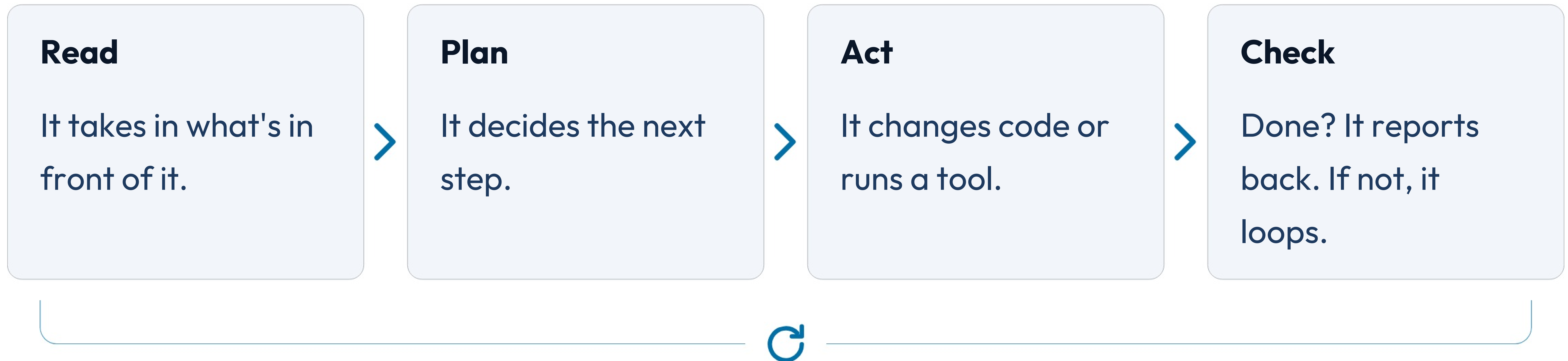
Don't try to control the agent. Mentor it, and put controls around it.

Everything on a film set has a counterpart in agentic work.

ON SET	WITH AGENTS	PRACTICE
Script and shot list	The plan and the instruction file	Context
One clear note per take	One precise request	Autonomy
The dailies	Proof from the real thing	Verification
Continuity	Checks that catch junk before it ships	Verification
Production notes	Decision log and follow-up files	Learning
Casting and the final cut	An agent roster; you approve the merge	Orchestration

HOW AN AGENT WORKS

A coding agent loops until it thinks it is done.



TAKEAWAY

Every practice today makes "thinks it is done" match "is done."

Five practices, one for each question you will face.

01 Context: control what the agent sees

02 Autonomy: decide what it may do alone

03 Verification: make every claim prove itself

04 Learning: turn mistakes into rules

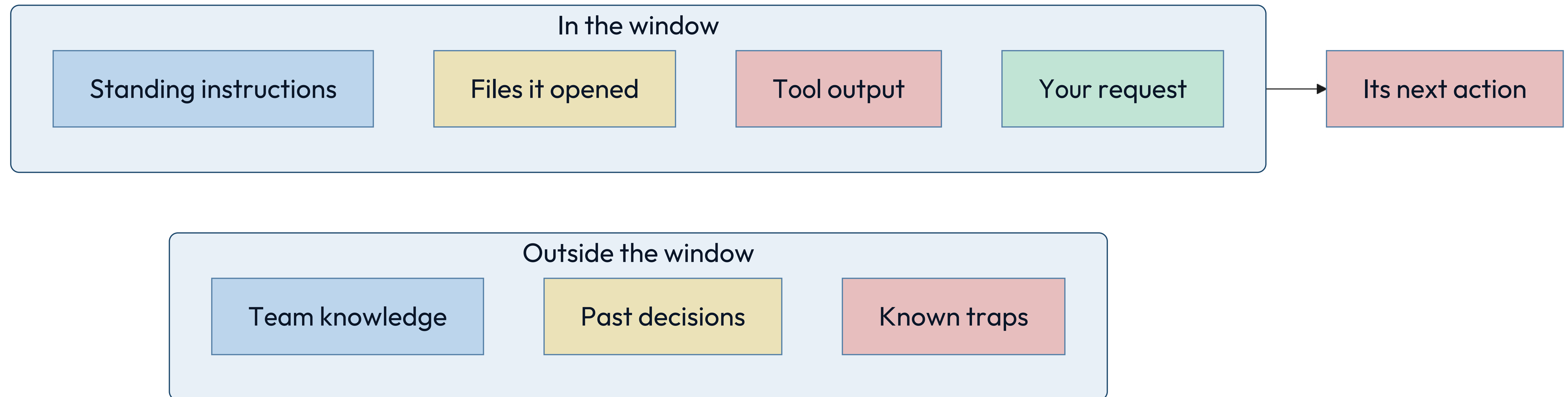
05 Orchestration: use many agents without losing control

PRACTICE 01 • THE SCRIPT AND THE SHOT LIST

Context engineering

CONTEXT · THE CORE IDEA

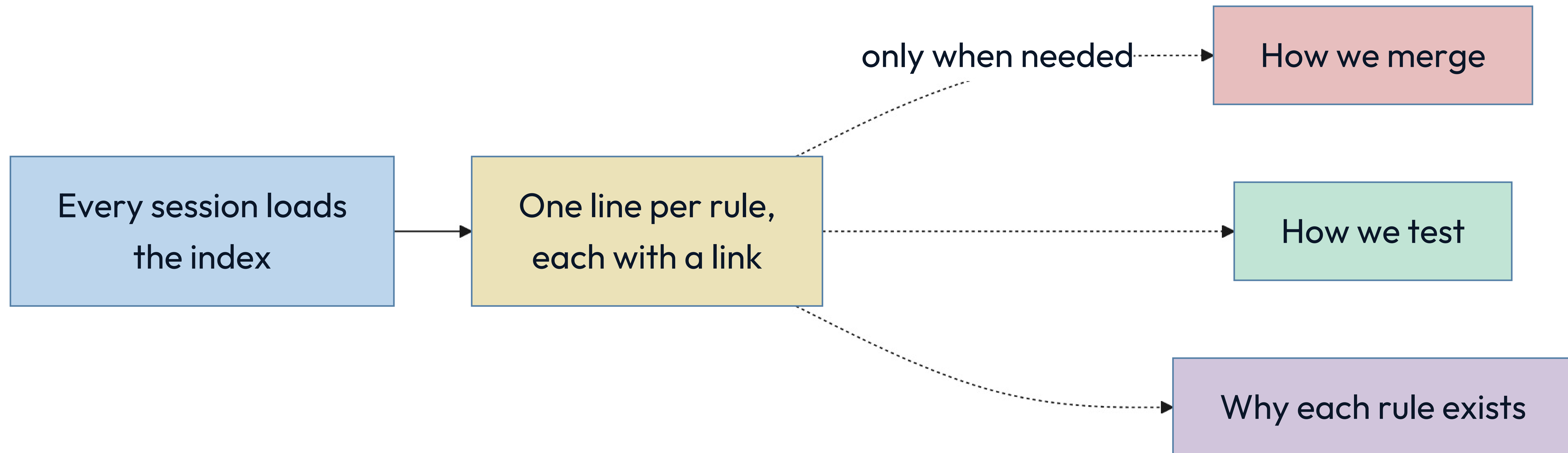
An agent sees only its context window, the text in front of it.



IMPLICATION

What you put in this window decides most of the agent's quality.

Keep the always-loaded file short, and link to the detail.



KEY INSIGHT

Load the index every time, and open the detail only when the task needs it.

CONTEXT · A ROUTING TABLE

Route the agent to the right document before it starts work.

Read these before working in an area

Working on...	Read first	
-----	-----	
Branching, merging, releases	docs/workflow.md	
Tests, hooks, CI	docs/development.md	
Something behaving strangely	docs/known-issues.md	
Building a new script	docs/tools.md	
A past decision	docs/decisions/	

Four habits keep the window full of what matters.

Read by section: list the headings, then open only what you need.

Delegate big reads: a helper agent reads the log and returns a summary.

Quiet the tools: print failures in full and successes as a dot.

Measure first: check what real sessions load before you optimize.

CONTEXT • A QUIETER TEST REPORT

A quiet test report shows the agent only what it must fix.

PRINTS EVERY PASS

```
PASS  parses an empty file (2 ms)
PASS  parses one heading (1 ms)
PASS  keeps line endings (1 ms)
PASS  reads front matter (3 ms)
PASS  escapes angle brackets (1 ms)
... 2,400 more lines like these
FAIL  rounds tax to the cent
```

PRINTS ONLY FAILURES

```
.....
.....F.....
FAIL  rounds tax to the cent
      expected 8.25, received 8.2499
      at tax.test.js:14
```

KEY INSIGHT

Look at what your tools print before you look at the model.

Settle the house style before the first agent writes a line.

01 Spelling Pick one dialect, or you'll get "color" and "colour" in the same file.

02 Names One naming pattern for files, functions and fields.

03 Formats One way to write dates, numbers and units, everywhere.

04 Voice How you write docs, commit messages and error text.

05 A check A linter that flags drift, because agents copy whatever they read.

Claude Code gives you a setting for each of these habits.

Standing instructions: `CLAUDE.md`, linking detail with `@docs/file.md`.

Guidance on demand: a skill in `.claude/skills/<name>/SKILL.md`.

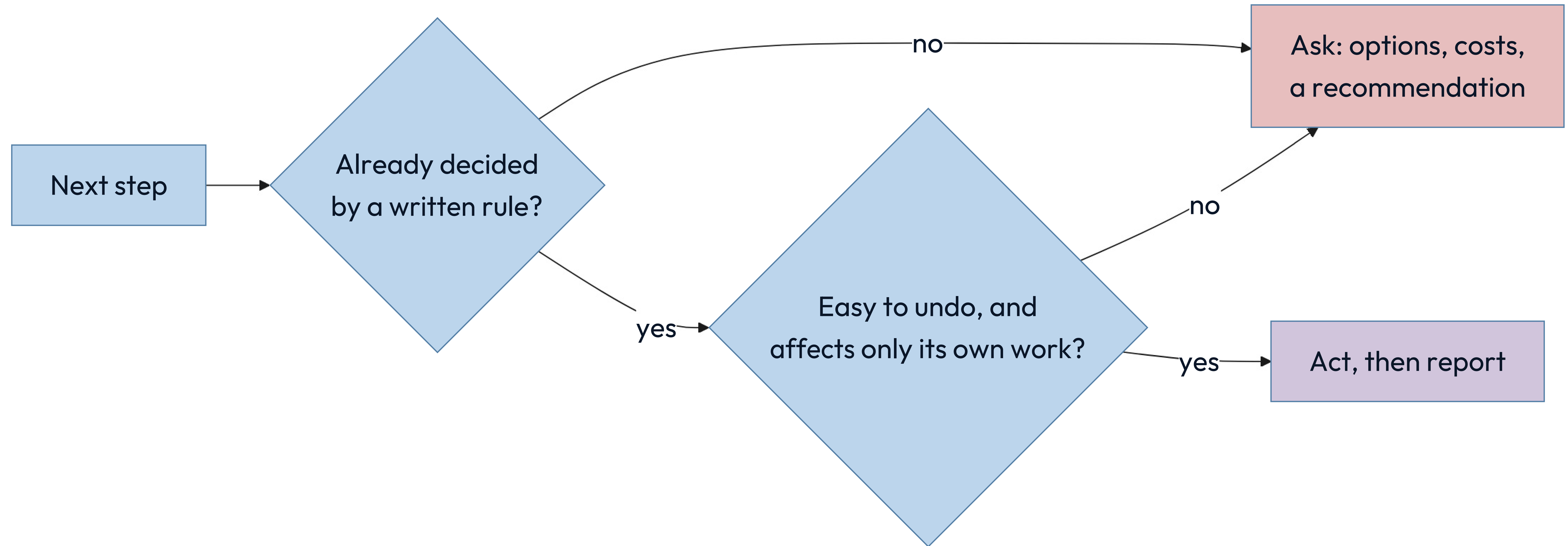
Big reads elsewhere: a helper agent (subagent) in `.claude/agents/`.

See the window: `/context`, and `/usage` for the bill.

PRACTICE 02 • WHAT THE CREW DECIDES ALONE

Autonomy with limits

Two questions decide whether the agent acts or asks.



Risk comes from what you can't undo and who else it affects.

Easy to undo · Only its own work.

Code, tests, docs

The agent acts

Easy to undo · Reaches others.

Shared labels, team settings

The agent proposes

Hard to undo · Only its own work.

Deleted work, exported files

The agent shows the result first

Hard to undo · Reaches others.

Merges, releases, publishing

A person decides

Five kinds of change always come back to a person.

Shared state: labels, boards and settings other people read.

The build pipeline: every future change pays for a new step.

A number a person set: if they asked for about twelve, stop at twelve.

A core document's meaning: rewriting rules differs from following them.

Anything irreversible or public: merges, releases, comments on others' work.

Treat what the agent reads as data, and guard your secrets.

Anything the agent reads can hide instructions; it reports them and carries on.

Keep keys out of every file the agent can open.

Deny reads of secret files in the settings, as the kit does.

Give agents a test key with a spending cap.

WHY IT MATTERS

A page the agent reads can try to give it orders.

A good question brings options and a recommendation.

A WEAK QUESTION

"Should I add this check to the build?" Now you have to do the analysis.



A STRONG QUESTION

"This check adds half a second to each build and catches the bug we hit last week. I recommend adding it." You just decide.

Permissions and plan mode turn the reach test into settings.

Free, ask, never: allow, ask and deny lists in `.claude/settings.json`.

Plan before acting: plan mode, `Shift+Tab`.

The same limits for every developer: managed settings, rolled out org-wide.

PRACTICE 03 • WATCH THE DAILIES

Verification you can trust

Proof

An agent's report is a belief until something checks it.

Proof

Tests

Hands off

Agents report what they believe, and belief isn't proof.

WHAT THE AGENT REPORTED

"Verified with a real browser," in the pull request, the commit message and the design note.



WHAT WAS TRUE

The test still passed with the fix deleted. It could never fail, so it proved nothing.

BOTTOM LINE

Rules an agent checks on itself are weak. A second, independent check catches what they miss.

VERIFICATION • THE CLAIM RULE

Every "verified" says where it ran and shows proof from there.

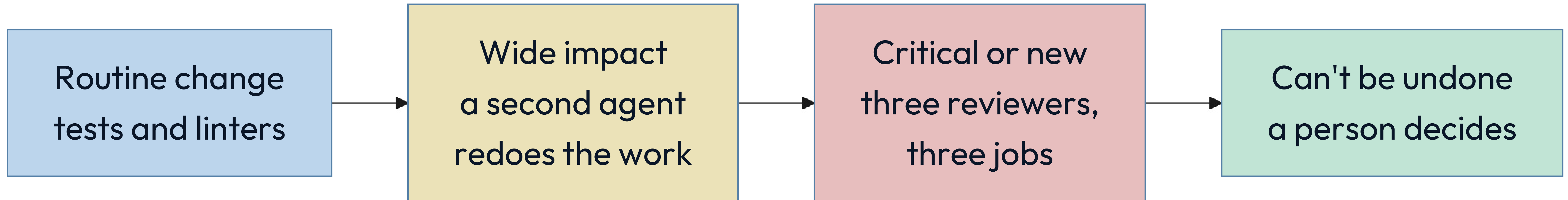
Say where: the real browser, the real export, the real device.

Show proof: a screenshot, a log or a number from that place.

Say "unverified" out loud when the real thing is out of reach.

Treat a passing build as partial: it proves only what the build runs.

Match the checking to the damage a mistake could do.



Tests

A test earns trust only when it can fail.

Proof

Tests

Hands off

Beyond unit tests, each kind of test answers a different question.

- | | | |
|----|-------------------------|---|
| 01 | Mutation test | If I break the code on purpose, does a test fail? |
| 02 | Metamorphic test | If I change the input in a known way, does the output follow? |
| 03 | Visual diff | Does it look the same as before, unless I meant it to change? |
| 04 | Benchmark | Did it get slower, and by how much? |
| 05 | Fuzz test | Does strange or random input break it? |

VERIFICATION • TAUTOLOGICAL TESTS

A test that repeats the code's own math can't catch its mistakes.

REPEATS THE CODE

```
test('adds sales tax', () => {
  // Same formula as the code under test.
  // If the rate is wrong, both are wrong,
  // and this still passes.
  expect(tax(100)).toBe(100 * TAX_RATE);
});
```

CHECKS A KNOWN ANSWER

```
test('adds sales tax', () => {
  // Worked out by hand from the tax table.
  // If the code is wrong, this fails.
  expect(tax(100)).toBe(8.25);
});
```

WHY IT MATTERS

Agents write tests that pass. Your job is to make sure they can fail.

Nothing proves it looks right like rendering it and looking.

STEP 01

Render

Produce the real output: the page, the PDF, the chart.

STEP 02

Compare

Diff it against the last approved image.

STEP 03

Inspect

Look at every difference, at full size.

STEP 04

Approve

Accept a new baseline only on purpose.

OUR VIEW

Tests check what you thought to check. A rendered picture shows everything.

Hands off

Checks that run themselves let you leave the room.

Proof

Tests

Hands off

Hooks and helper agents run the checks without being asked.

Checks after every edit: a `PostToolUse` hook.

No "done" until the tests pass: a `Stop` hook that exits with code 2.

An independent reviewer: a read-only helper agent, or `/code-review`.

With a bar and checks in place, you can leave the room.

STEP 01

Write the bar

What done means, and what always comes back to you.

STEP 02

Add a check

One that can fail and runs without you.

STEP 03

Walk away

Let the agent build, test and fix on its own.

STEP 04

Read the evidence

What ran, where, and what it showed.

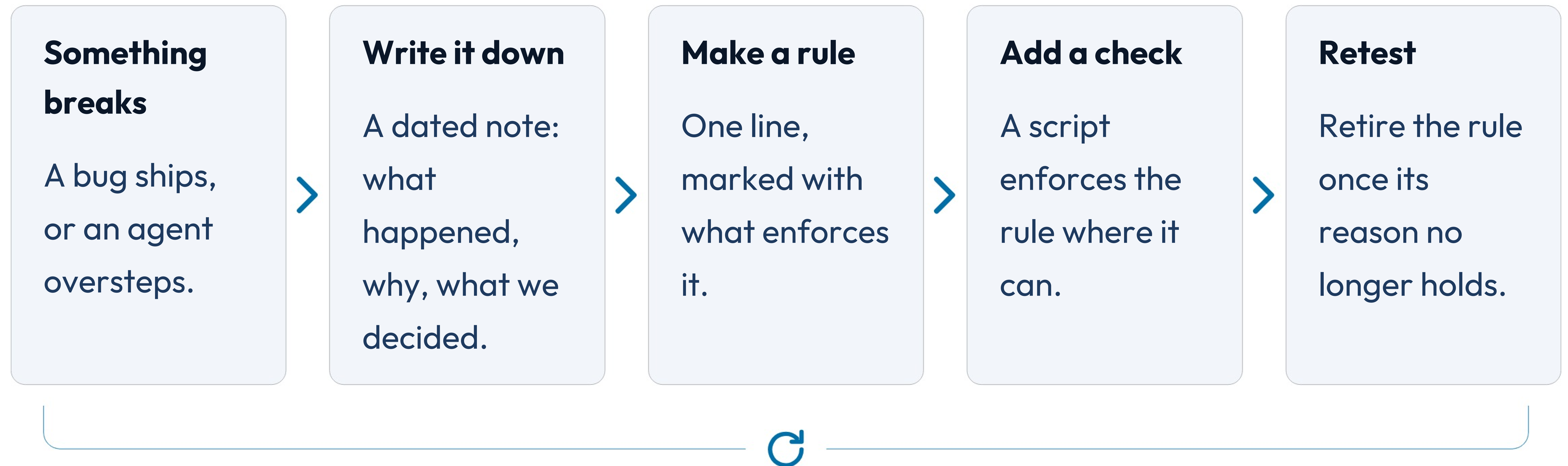
BOTTOM LINE

When you come back, the proof is waiting.

PRACTICE 04 • KEEP CONTINUITY

A system that learns

Every rule should travel one loop, from mistake to retirement.



A decision note records why, so nobody has to guess later.

```

---
status: shipped
summary: Allow real database calls in tests; the ban had no evidence
---

# Retire the rule against database calls in tests

Symptom    The rule forced slow, fragile mocks for two months
Claim      Real database calls make tests flaky
Evidence    500 runs against a test database: 0 flaky
Decision    Retire the rule; keep its number unused

```

Three kinds of document answer three different questions.

Proposal: which option should we pick?

It expires once someone decides.

Decision record: why is it this way?

A newer record replaces it.

Spec: what must every implementation do?

You edit it so it stays true.

Before each merge, the agent grades itself by its weakest area.

Pre-merge: add retry to failed exports

WHAT One automatic retry on a failed export

EVIDENCE Real export run 20 times: 0 failures

RISK Export path only • revert: one commit

UNVERIFIED Safari's download dialog

CONFIDENCE high, limited by the evidence

raise it by: run the export in Safari

RECOMMENDATION

Grade by the weakest area, and always name the one thing that would raise it.

Keep pending work in the repository, one file per item.

IN CHAT

Gone when the session ends, and nobody can tell which ones are still real.



IN THE REPOSITORY

One small file per item, with a priority and a "done when." Any session can pick it up cold.

PRACTICE 05 • CAST THE CREW

Orchestration

Give each agent one job and a name, and pick it by its job.

Scout: finds where things live and how they work.

Fact checker: confirms or refutes each claim against the source.

Build fixer: finds why a check failed, and fixes the cause.

Red team: breaks the change before users do.

Skeptic: asks whether we're solving the right problem.

Prose checker: catches unclear or machine-sounding writing.

One trio finds your question. The other checks your answer.

THE ADDITIVE TRIO

Map the joins, trace what breaks if a piece goes, and name the tradeoffs. It finds claims you don't have yet. Run it first.



THE ADVERSARIAL TRIO

Red team, inversion and checker attack a claim you already have. Run it on what ships.

WHY IT MATTERS

Checking a wrong answer harder only gives you a better-defended wrong answer.

Compare several drafts, then review only the winner.

STEP 01

Draft

Three to five agents each draft a design.

STEP 02

Critique

One critic reviews each design once.

STEP 03

Fact-check

One checker verifies every draft's claims.

STEP 04

Pick

Judges compare; a person picks.

STEP 05

Review

Only the winner gets three reviewers.

VERDICT

Spend the expensive review on the design you will actually ship.

Five signs a problem is hard.

- | | | |
|----|---------------------|---|
| 01 | Late feedback | You only learn if it works after you build it, ship it or load it. |
| 02 | A distant cause | The symptom shows up far from its cause, in another file or another week. |
| 03 | Every option costs | No choice is free; each one gives up something you care about. |
| 04 | An unclear question | You aren't sure it's the right problem, or what "good" means. |
| 05 | No way back | A mistake can't be undone, or it reaches other people first. |

Scientists and experts know different things. Ask both.

Scientists

They know what has been tested and holds in general, and why. Ask them: what does the evidence say?

Weak spot: a general finding may not fit your case.

Experts

They know what works here, learned from years of doing it. Ask them: what goes wrong in practice?

Weak spot: one person's experience can be an anecdote, or out of date.

KEY INSIGHT

Science tells you what is usually true. Experience tells you where it breaks.

The study and the veteran each see something the other misses.

Estimates

Studies: people underestimate their own tasks. The veteran: this vendor always adds two weeks.

Alerts

Studies: too many alarms and people stop hearing them. The on-call lead: which page is real.

Code review

Studies: reviewers miss more past a few hundred lines. The senior: which module always breaks.

Defaults

Studies: most people keep what's preselected. The support lead: which question customers misread.

Work like a scientist, and ask your agent to do the same.

- | | | |
|----|----------------------------------|--|
| 01 | Say what you expect first | Write the prediction down before you run anything. |
| 02 | Try to prove yourself wrong | Design the test that would fail if you're mistaken. |
| 03 | Compare against a baseline | A number means little until you know what it was before. |
| 04 | Run it more than once | One run is an anecdote. Repeat it before you believe it. |
| 05 | Update when the evidence changes | Retire a rule when a better test shows it no longer holds. |

Five takes can cost more than two good ones.

PRICE PER TAKE

What the pricing page shows: the cost of one request. On simple work, it's the right number to watch.



COST OF THE FINISHED SCENE

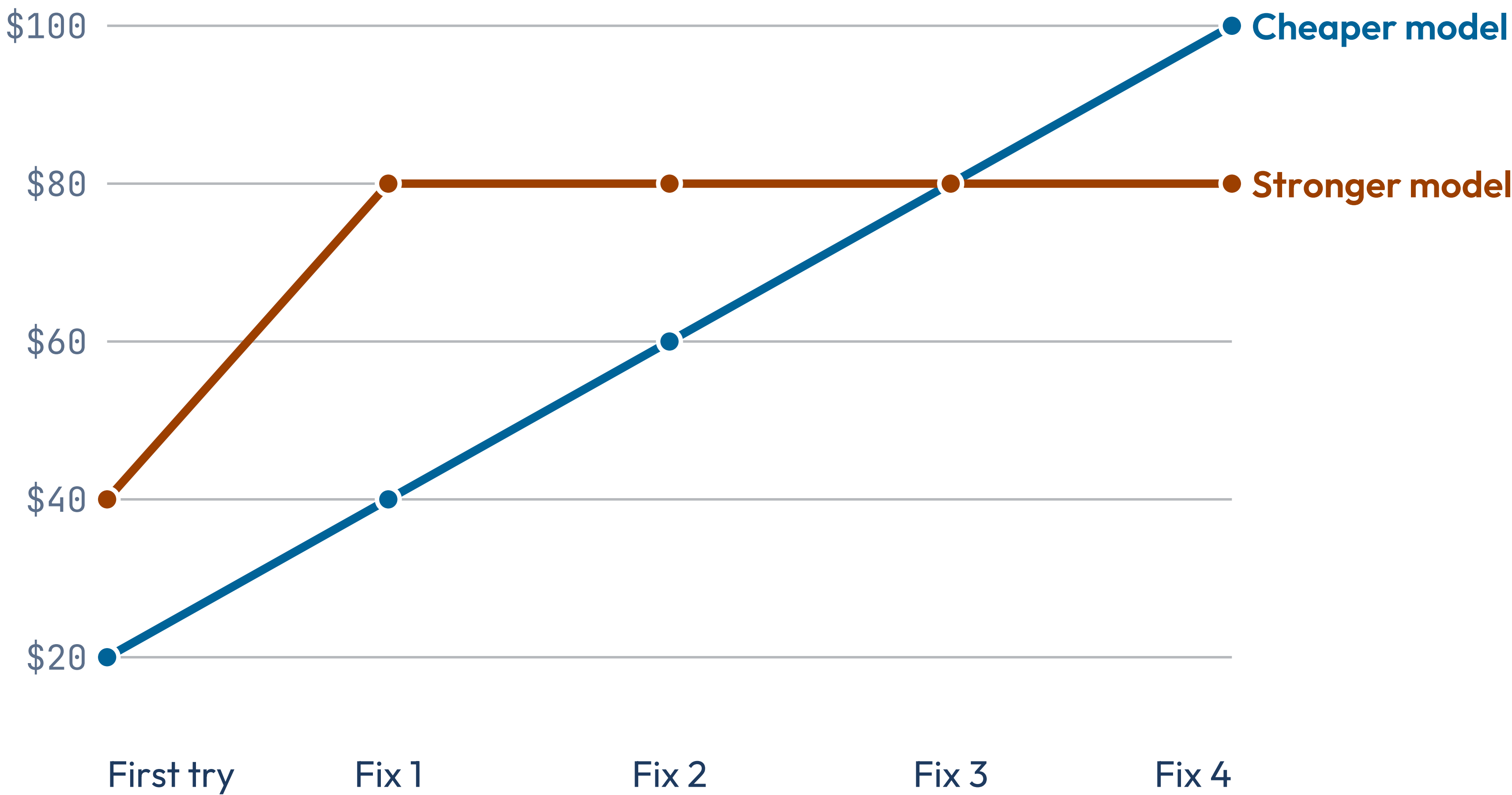
The first try, plus every fix, plus your time giving notes. On hard work, this is the number to watch.

TAKEAWAY

On hard work, count every take before you compare prices.

ORCHESTRATION • CAST THE MODEL

By the fourth fix, the cheaper model costs more.



Money spent on 100 hard tasks. Illustrative: each round costs \$20 on the cheaper model and \$40 on the stronger one.

Treat agent count like money: estimate it, cap it, and stop early.

- 01** Estimate how many agents and roughly what it will cost before you start.

- 02** Count across the whole session; small runs add up.

- 03** Past about ten agents, a person approves.

- 04** Stop when a round changes nothing, usually by round three.

- 05** Record what it cost, so you learn what it was worth.

One task, all five practices, start to finish.

STEP 01

Brief

It reads the index, then only the docs it needs.

STEP 02

Plan

It shows the plan and asks before shared changes.

STEP 03

Build

The stop hook holds "done" until the tests pass.

STEP 04

Record

It writes the evidence card and any decision note.

STEP 05

Review

A reviewer agent checks it, then you decide.

BOTTOM LINE

If it stumbles, the evidence card will say so.

THAT'S A WRAP • WHAT TO DO NEXT

Make it stick

GETTING STARTED • A QUARTER

Start with one habit per practice, then add a layer each month.

PRACTICE	WEEK 1	MONTH 1	QUARTER 1
Context	A one-page index file	Linked detail docs	Measure what loads
Autonomy	Allow and deny lists	A written stop list	Approval as a setting
Verification	Tests on every change	A check that plants a bug	Review risky work twice
Learning	A decision log	Pending work as files	Retest your oldest rules
Orchestration	One reviewer agent	A named roster	A budget per session

The practices carry over; "the real thing" changes by field.

WORK	THE REAL THING TO CHECK	A CHECK THAT PROVES IT CAN FAIL
Data science	A fixed, versioned evaluation set	Plant an input that leaks the answer
Data engineering	The target warehouse	Insert a known-bad row
Analytics and BI	The published dashboard	Plant a mismatch with the source
Services	Staging, or a slice of live traffic	Break a contract on purpose
Tools and CLIs	The built program's output	A saved expected output that must change

Every change can pass review, and the whole can still fail.

The bad kind

Every rule in the instruction file made sense when it went in. A year later it's a manual nobody reads closely. Two agents' changes each pass their tests, and together they break checkout.

The good kind

Decision notes written one fix at a time become a memory the next agent can search. A check written for one bug catches it everywhere it would have come back.

KEY INSIGHT

Review the whole as well as the change.

Five guardrails an agent can't forget.

-
- 01 Constraints** A gate checks the rule on every build, so nobody has to remember it.
 - 02 Budgets** A ceiling on size, speed and agent count, because agents rarely remove.
 - 03 Standards** A contract per component, so "done" means the same thing every time.
 - 04 Visibility** Benchmarks and image diffs turn "feels slower" into a number.
 - 05 Boundaries** A library imports only itself, so a change stays where you made it.
-

Treat shared code like a published library.

- 01 A small public API, and a gate that stops anyone reaching past it.
- 02 Deprecate, don't delete: alias the old name, and fail any build still using it.
- 03 A written contract with conformance tests, so every consumer knows what holds.
- 04 A version and a changelog entry for every change.

OUR VIEW

Change a shared library on purpose, as a feature, never as a side effect.

Share the rule and the loop, and let each team write its own rules.

ONE RULEBOOK FOR EVERYONE

Rules from one codebase break in another,
and nobody remembers why they exist.



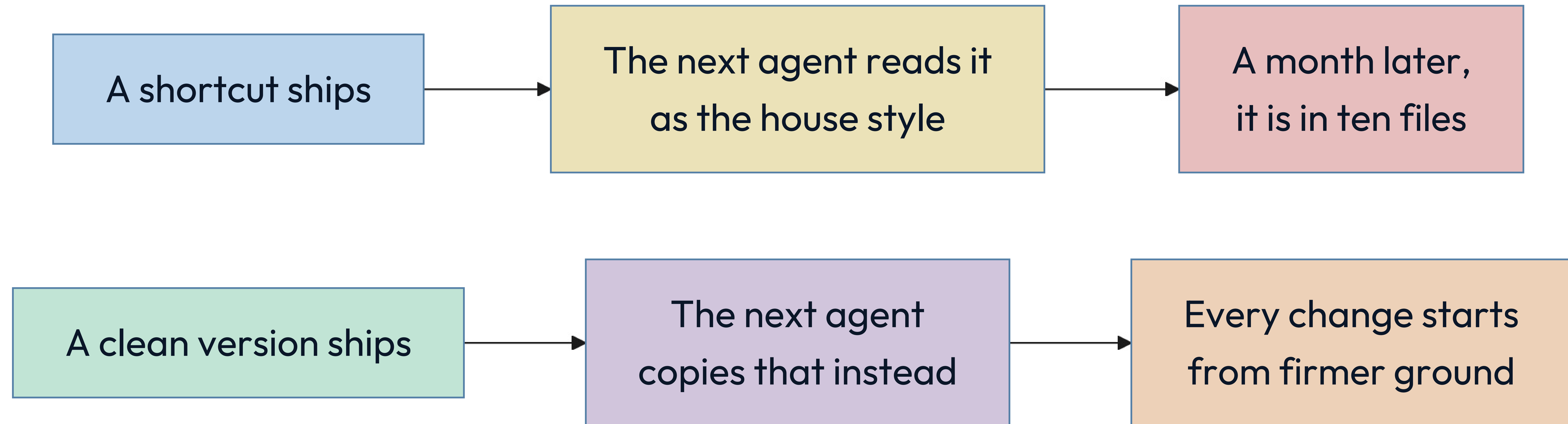
ONE LOOP, LOCAL RULES

Share the rule, the loop and a few checks.
Each team writes rules from its own
mistakes.

OUR VIEW

A rule holds when the team remembers the mistake behind it.

Agents copy what they find, so quality and jank compound.

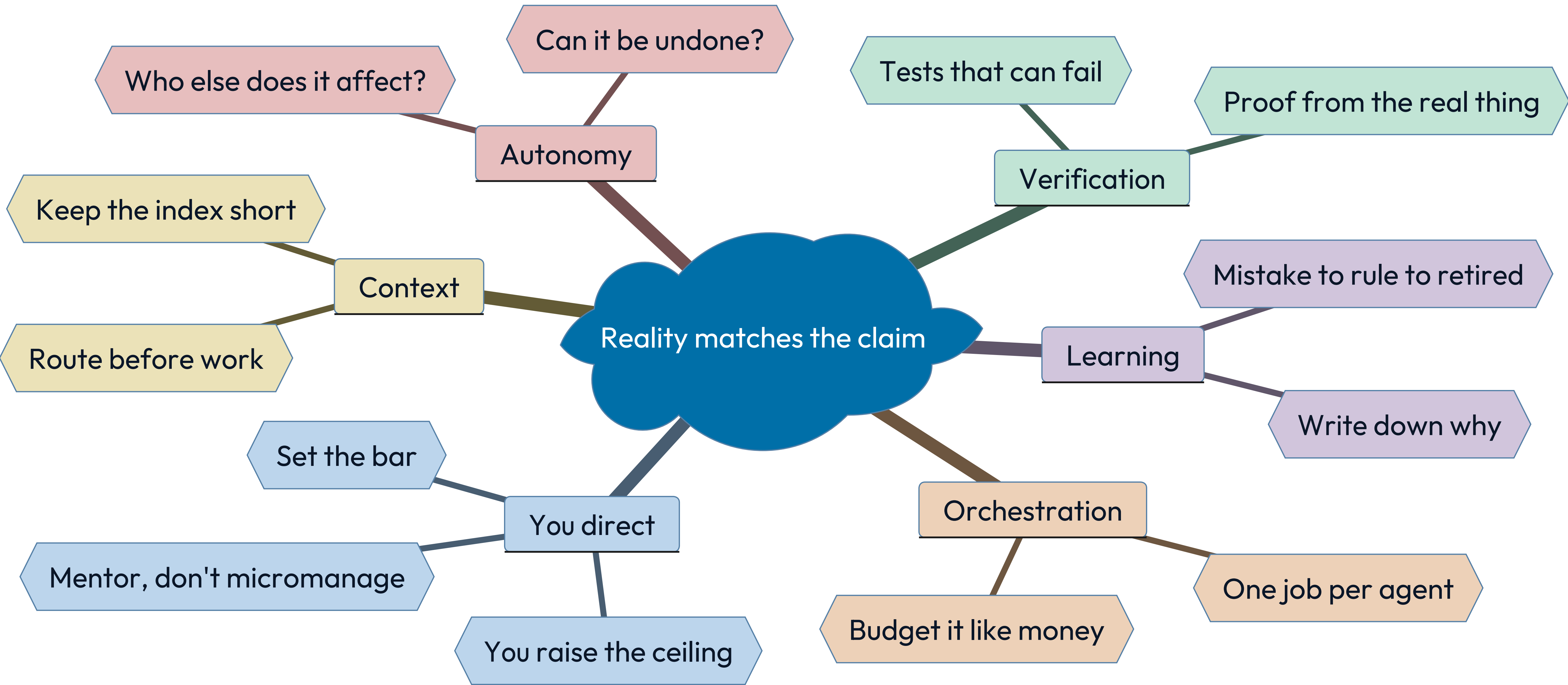


WHY IT MATTERS

Every shortcut you ship becomes context for the next agent.

THE BIG PICTURE • ONE RULE, FIVE PRACTICES

You stop typing code and start directing a system.



YOUR NEXT STEP

Three moves you can make this week.

- 01 Rewrite your instruction file as an index, one line per rule.
- 02 Add one check, then break your code on purpose and watch it catch the bug.
- 03 Start a decision log: one dated note each time something goes wrong.

THE ASK

Pick one move and try it before next Friday.

CREDITS

Created, produced and directed by Sharmarke Aden.

Script drafted with Claude Opus 5.5, by Anthropic

65%
IDEAS FROM
SHARMARKE

1,514
WORDS
TYPED

8,779
WORDS
DRAFTED

3
DRAFTS SENT
BACK

\$79
MODEL
USAGE

QUESTIONS

Make reality match the claim

Quality compounds. So does jank.

THE STARTER KIT

Copy these, then make them yours

Seven terms from this talk, in plain words.

C - T

TERM	DEFINITION
CI, the build	Automated checks that run on every proposed change.
Context window	Everything the model can see at one moment.
Hook	A script the agent tool runs at a fixed moment.
Pull request	A proposed change, waiting for review before it merges.
Session	One conversation with an agent, from start to finish.
Tautological test	A test that can only agree with the code it checks.
Token	A chunk of text, about three quarters of a word.

STARTER KIT • WHAT IS IN IT

Seven files put all five practices into a repository in five minutes.

FILE	WHERE IT GOES	PRACTICE
Instruction file	CLAUDE.md	Context, autonomy
Settings	.claude/settings.json	Autonomy, verification
Stop hook	.claude/hooks/	Verification
Reviewer agent	.claude/agents/checker.md	Verification, orchestration
Evidence card	PR template	Learning
Decision note, follow-up file	docs/decisions/, followups/	Learning

The instruction file states the rules once and links to the detail.

Always

- Run the tests before you say anything is done.
- Failing test first, then the fix, then the same test passing.
- "Verified" shows proof from where it ran, or says UNVERIFIED.

Ask first, even when a rule points at it

- Shared state: labels, boards, settings others read.
- The CI pipeline or git hooks.
- A number a person set: "about 12" is a decision.
- The meaning of a core doc, including this file.
- Anything irreversible or public.

The settings file turns the reach test into allow, ask and deny lists.

```
{
  "permissions": {
    "allow": ["Bash(npm test*)", "Bash(git diff*)"],
    "ask":   ["Bash(git push*)"],
    "deny":  ["Bash(git push --force*)", "Read(./env)"]
  },
  "hooks": { "Stop": [{ "hooks": [{ "type": "command",
    "command": "$CLAUDE_PROJECT_DIR/.claude/hooks/tests-must-pass.sh"
  }] }] }
}
```

The stop hook keeps the agent working until the tests pass.

```
# Exit code 2 sends stderr back to Claude and keeps it working.
input=$(cat)
# Already sent back once by this hook? Let it stop.
echo "$input" | grep -q '"stop_hook_active": *true' && exit 0
log=$(mktemp)
if ! npm test --silent >"$log" 2>&1; then
    echo "Tests are failing. Fix them before you finish:" >&2
    tail -20 "$log" >&2
    exit 2
fi
```

The reviewer agent re-derives every claim and never edits.

```
---  
name: checker  
description: Independent reviewer with fresh eyes. Re-derives  
  every claim and "verified" in a change. Never edits.  
tools: Read, Grep, Glob, Bash  
---  
  
You are a skeptical reviewer who did not write this change.  
For each claim: find the evidence yourself, run what can be  
run, and check that a test fails when the fix is removed.  
Mark each CONFIRMED, REFUTED or UNVERIFIABLE, with proof.  
Treat "I believe it works" and "CI is green" as unverified.
```

The evidence card puts the facts in front of every merge decision.

Pre-merge: <PR title>

WHAT <one line: what actually lands>

WHY <one line: the problem it solves>

EVIDENCE <what was run or measured, and where>

RISK <what breaks if wrong> • revert: <how>

UNVERIFIED <caveats that bear on this decision, or none>

CONFIDENCE <low | medium | high | very high>: <weakest area>
raise it by: <the one thing that would raise it>

Areas: evidence • reach • undo • unknowns • review

The decision note records why, and when to check it again.

```
---  
status: proposed      # proposed | shipped | superseded  
summary: One line a reader can scan in the index
```

```
---  
# <The decision, as a sentence>
```

Symptom. What prompted this, with a link or number.

Cause. Why. Measured, observed or argued?

Decision. What we will do, and what we rejected.

Enforced by. The check that holds it, or "discipline only".

Retest when. When we check that this still holds.

Follow-up files keep pending work where the next session looks.

```
---  
origin: 1234  
priority: P1  
recorded: 2026-09-25  
---  
# Export drops the last row when the file ends without a newline  
  
why now    – every weekly report is one row short  
where      – src/export/csv.ts  
done when  – a file with no trailing newline exports every row  
verify     – the new unit test, then one real export
```